DSA Experiment 9

Title: Represent graphs and generate minimum spanning trees using Prim's algorithm.

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CODE:

```
#include <stdio.h>

#define V 5

int main()
{
   int graph[V][V] = {
      {0, 2, 0, 6, 0},
      {2, 0, 3, 8, 5},
      {0, 3, 0, 0, 7},
      {6, 8, 0, 0, 9},
      {0, 5, 7, 9, 0}
   };

   int visited[V] = {0};
   int edges = 0;
   int min, u, v, total = 0;

   visited[0] = 1;  // Start from vertex 0

   printf("Edge  Weight\n");

   while (edges < V - 1)
   {
      min = 999;

      for (int i = 0; i < V; i++)
      {
         if (visited[i] == 1)
         {
            for (int j = 0; j < V; j++)
            {
               if (visited[j] == 0 && graph[i][j] != 0)
               {
                  if (graph[i][j] < min)
                  {
                     min = graph[i][j];
                     u = i;
                     v = j;
                  }
               }
            }
```

```
            }
        }

        printf("%d - %d   %d\n", u, v, min);

        visited[v] = 1;
        total = total + min;
        edges++;
    }

    printf("Total MST Weight = %d\n", total);

    return 0;
}
```

Output:

